

Masking and Temporal Niche Switches in Spiny Mice

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Abstract Activity patterns are the product of interactions between an internal circadian clock and direct responses to photic and nonphotic features of the environment that are said to “mask” the influence of that clock. Evolutionary transitions between nocturnality and diurnality involve changes in mechanisms underlying both of these processes. Here, the authors examined how masking influences activity patterns of golden spiny mice (*Acomys russatus*), which can be either nocturnal or diurnal, and common spiny mice (*Acomys cahirinus*), which are strictly nocturnal. Animals kept on a 12:12 LD cycle were exposed to 3-h dark pulses starting at ZT 2, light pulses of varying intensities (50, 100, 700, or 1500 lux) at ZT 14, and a 3.5:3.5-h LD cycle. In common spiny mice, activity increased by 379% during the dark pulse and decreased during light pulses to 23% of baseline levels. Golden spiny mice also increased their activity in response to the dark pulse (by 345%), but there was extreme inter- and intraindividual variability and no significant response to light pulses at night. In the 3.5:3.5 LD cycle, common spiny mice showed a preference for the dark phase with $86\% \pm 0.01\%$ of activity occurring then, whereas golden spiny mice showed a pronounced circadian rhythm but no evidence of masking. Masking responses to light and dark were thus unsurprising in common spiny mice but were highly unusual in golden spiny mice. Patterns seen in the latter species may reflect mechanisms enabling these animals to occupy either a diurnal or a nocturnal niche in their natural habitat.

Key words masking, diurnal, nocturnal, spiny mice, temporal partitioning, activity pattern

The activity patterns of mammals are heavily influenced by light through entrainment of an internal circadian clock and through masking (Redlin et al., 2005). Whereas entrainment is quite similar in nocturnal and diurnal species, masking is very different. Specifically, light increases activity in diurnal mammals (positive masking) and suppresses it in nocturnal ones (negative masking), while darkness acts in the opposite way. In order for a nocturnal species to evolve into a diurnal one or for an individual to move from a nocturnal to a diurnal niche, the negative

masking effects of light on activity must be overcome. Golden spiny mice (*Acomys russatus*) represent a unique species with which to investigate such issues because they are diurnal in most, but not all, of their natural habitat and because they abruptly become nocturnal when transferred from the field into the lab. These animals possess a mosaic pattern of adaptations, some supporting diurnality (e.g., dark skin pigmentation) and others supporting nocturnality (e.g., some features of the structure of the retina), which has led to the suggestion that they represent an

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evolutionary transition between these 2 modes of adaptation to the day-night cycle (reviewed in Kronfeld-Schor and Dayan, 2008).

Golden spiny mice have a strictly nocturnal congener, the common spiny mouse (*Acomys cahirinus*), with which comparisons can be made to clearly identify traits that have enabled them to move from one temporal niche to another. Here, we directly compared golden and common spiny mice to determine whether their masking responses to light and dark reflect adaptations to support nocturnality, diurnality, or some combination of the two.

MATERIALS AND METHODS

Animals, Housing, and Data Collection

Animals were adult male and female golden and common spiny mice from our breeding colony at the I. Meir Segals Zoological Garden at Tel Aviv University. Golden spiny mice are consistently nocturnal in the lab, regardless of ambient temperature (absolute levels or daily changes) and housing conditions (indoors or outdoors) and even when housed in the same room or cage as common spiny mice. The source of the breeding colony is Ein Gedi, near the Dead Sea, Israel, where the 2 species coexist. The mice (each species in a separate room) were individually housed in 38 × 24 × 13-cm plastic cages and kept at 28 °C in a 12:12 LD cycle with standard rodent chow (Koffolk serial no. 19510, 21% protein, 4.5% fat, 19.3 kJ/g) and water available *ad lib*. Light intensity during the light phase of the LD cycle was approximately 800 lux. All procedures were conducted in accordance with and approved by the Institutional Animal Ethics Committee (L-02-45). All efforts were made to minimize the number of animals used and their discomfort. Activity was continuously recorded using an infrared detector (Intrusion detector model MH10, Crow group, Israel) connected to a PC. Data were collected at 6-min intervals using Strawberry tree software.

Experimental Design

Experiment 1: Dark Pulse

Twelve golden and 11 common spiny mice were housed in individual cages under 12:12 LD conditions. After a 3-week acclimation period, activity was

monitored continuously for 7 days using infrared detectors. On day 8, lights were turned off for 3 h at ZT 2 (ZT 0 = lights-on), after which the original LD cycle was reinstated. Activity was recorded for another 7 days.

Experiment 2: Light Pulses

Seven days after the dark pulse in experiment 1, the same animals were exposed to a 3-h light pulse beginning at ZT 14. This was repeated 4 times such that each animal was exposed to pulses of 4 different light intensities—50, 100, 700, and 1500 lux—with 1-week intervals between each pulse.

Experiment 3: 3.5:3.5 Ultradian LD Cycle

This experiment was performed 1 week after the last light pulse in experiment 2 on the same animals. Here, we used a 3.5:3.5 ultradian LD cycle to which the circadian rhythms cannot entrain; consequently, the animals are exposed to light and to darkness at all phases of their circadian rhythms in the course of 1 week (Redlin and Mrosovsky, 1999). All spiny mice were monitored for 2 weeks under the ultradian LD cycle, but only the second week was analyzed. The level of activity (counts per minute) during the dark and light phases of the LD cycles was determined for a 1-week period, and the percentage of the total that occurred at night was calculated.

Data Analysis

Data were analyzed using the Clocklab program (Actimetrics Wilmette, IL), which was also used to generate all actograms, and chi-square analysis. Microsoft Excel was used to generate the figures, and STATISTICA was used for all statistical tests. In experiment 1, absolute activity levels during the dark pulse were compared with activity level for the corresponding time the preceding day using paired *t* tests. In experiment 2, the effects of the pulses were examined using repeated measures 2-way analysis of variance (ANOVA) on absolute activity levels with both masking/no masking and light pulse intensity as repeated variables; Scheffé tests were used for post hoc comparisons. In experiment 3, masking scores were compared using a paired *t* test on absolute activity levels. Differences were considered significant if $p < 0.05$. Data are presented as mean ± SEM.

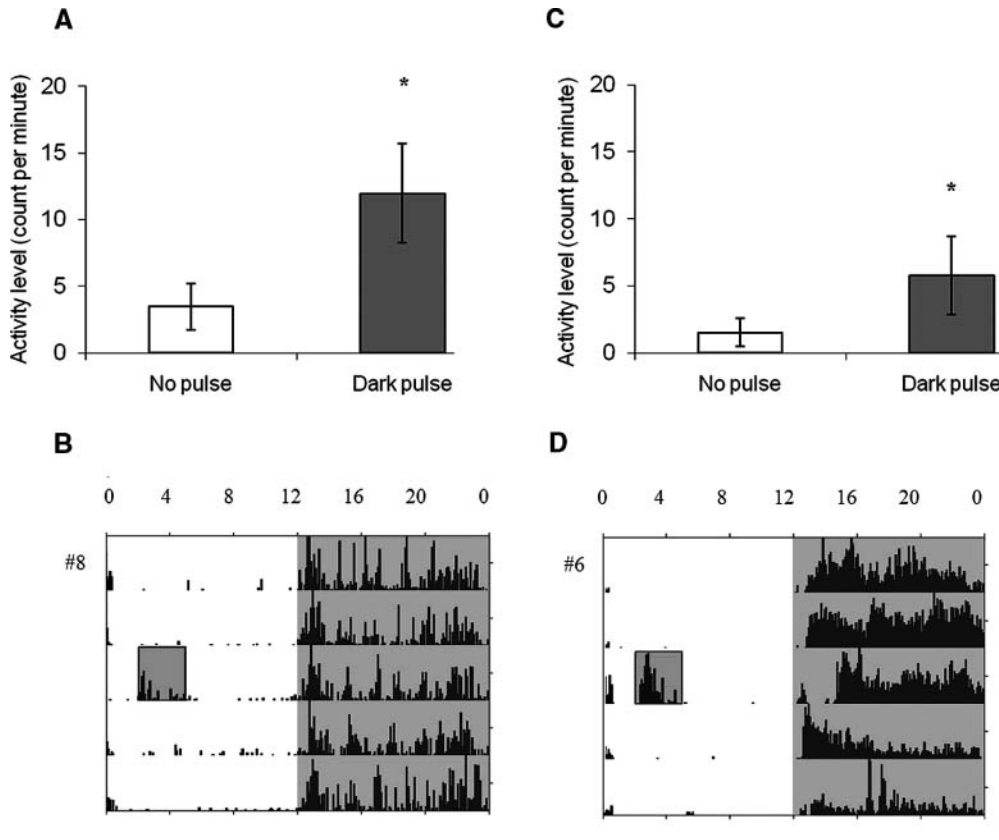


Figure 1. Activity levels of (A) common and (C) golden spiny mice (counts per minute, mean $\pm$ SE; $n = 11$ and 12 , respectively) with (gray) or without (white) 3 h of dark pulse starting at ZT 2, as well as a representative actogram for (B) common and (D) golden spiny mice kept in LD 12:12 h, showing the response to the 3-h dark pulse (ambit with square). * $p < 0.01$.

RESULTS

Experiment 1: Dark Pulse

The responses to the 3-h dark pulse that began at ZT 2 were similar in the 2 species, with all individuals showing positive masking. Activity during the dark pulse was $379\% \pm 97\%$ and $345\% \pm 169\%$ higher than at the same time the preceding day for common and golden spiny mice, respectively (Figs. 1A,C). Examples for each species are depicted in Figures 1B,D. These increases were statistically significant in each species (golden spiny mouse: $t(11) = -3.4$, $p < 0.01$; common spiny mouse: $t(10) = -4.1$, $p < 0.01$).

Experiment 2: Light Pulses

In common spiny mice, there was an effect of the pulse (repeated-measures ANOVA; pulse vs. no pulse; $F(1, 40) = 56.358$, $p < 0.0001$), no effect of light intensity ($F(3, 40) = 1.1138$, NS), and no interaction between these 2 variables ($F(3, 40) = 0.6296$, NS; Fig. 2A). In this species, activity levels during the light dropped to $23.75\% \pm 0.4\%$ of those seen during the

same hours the preceding day (Fig. 2A). In 5 individuals, activity remained suppressed for 0.5 to 1 h after the light pulse ended (e.g., Fig. 2B). We also found that in response to the light pulse, 7 individuals delayed their activity onset the next day (by 57.29 ± 4.72 min; e.g., Figs. 2B,C).

Among golden spiny mice, there was a great deal of both inter- and intraindividual variability with respect to responses to the light pulses (Figs. 2E-H), and there were no overall effects of the light pulses (Fig. 2E, pulse vs. no pulse: $F(1, 44) = 0.6036$, NS) or light intensity ($F(3, 44) = 0.6048$, NS) on activity levels, and there was no interaction between these

variables ($F(3, 44) = 0.8339$, NS). At each light intensity, activity during a light pulse relative to that at the same time the preceding day was lower in some individuals and higher in others, and in some cases there was no change at all (Table 1). These patterns did not simply reflect interindividual variability, as animals were not consistent with respect to how they responded to the light. On average, individuals decreased their activity in $52\% \pm 12.18\%$ of the light pulses, did not change it in $21\% \pm 7.39\%$, and increased it in $27\% \pm 9.57\%$ of the light pulses. Only 1 individual responded consistently to the 4 light pulses (decreasing its activity in response to all of them). We also found no correlation between the ratio of day/night activity of animals on the day before the light pulse and the response of that animal to the light pulse; that is, a more nocturnal pattern of activity did not predict a greater inhibitory response to light.

Experiment 3: 3.5:3.5 LD Cycle

In common spiny mice, $86\% \pm 0.01\%$ of activity occurred during the dark phase of the 3.5:3.5 LD cycle

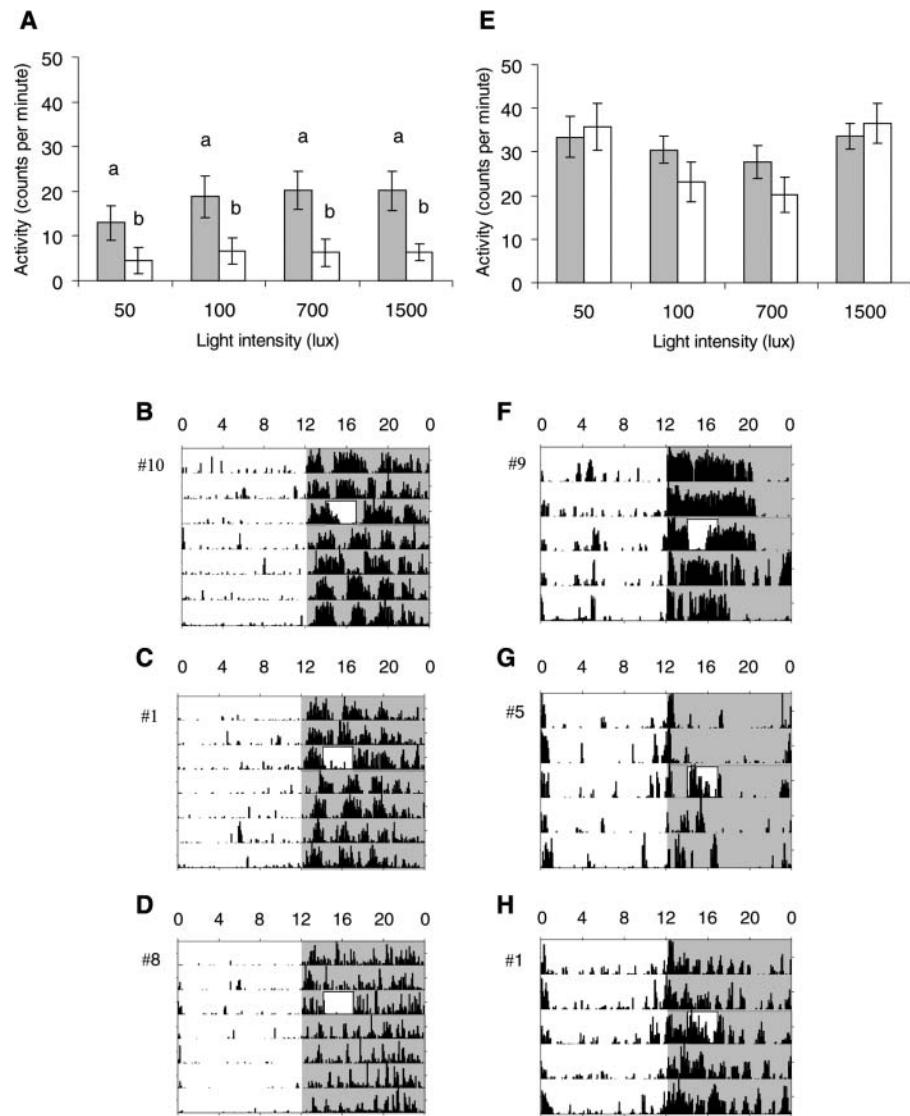


Figure 2. Activity levels (counts per minute, mean $\pm$ SE) of (A) common ($n = 11$) and (E) golden ($n = 12$) spiny mice during exposure to 3 h of light pulse in 4 different irradiances (50, 100, 700, and 1500 lux, white bars) starting at ZT 14 and during the same hours the night before (gray bars), as well as representative actograms for (B–D) 3 common spiny mice and (F–H) 3 golden spiny mice kept in LD 12:12 h and exposed to the 3-h light pulse (ambit with square). All common spiny mice reduced activity level in response to the light pulse, and some showed a delayed activity onset the day after the light pulse (e.g., B). The response of golden spiny mice to the light pulses was more variable, with some decreasing activity level (e.g., F), some increasing activity level during the light pulse (G), and some not responding to it at all (e.g., H). Bars with different letters are significantly different from each other ($p < 0.0001$).

Table 1. Responses to 3-h Light Pulses between ZT 14 and ZT 17

	Decreased	No Change	Increased
50 lux	7	4	1
100 lux	6	0	5
700 lux	6	1	5
1500 lux	6	3	3

Numbers of individuals in which activity during the 3-h light pulse declined by more than 20%, remained the same, or increased by more than 20%, relative to levels seen during the same interval the preceding day.

(Figs. 3A, B1,C1,D1), which was significantly more than during the light phase ($t(10) = 8.3$, $p < 0.0001$). Several significant peaks with periods that were multiples of 3.5 (e.g., 14, 21, 28, 35 h) as well as one of 24 h were found in common spiny mice (Figs. 3B2, C2,D2). By contrast, in golden spiny mice, 50% $\pm$ 0.02% of activity occurred in the dark phase (Figs. 3E,F1,G1, H1), and light had no effect on activity ($t(11) = 0.06$, NS; Figs. 3E,F1,G1, H1). Chi-square analysis revealed only one significant peak, at 24.6 $\pm$ 0.3 h, in golden spiny mice (Figs. 3F2,G2,H2).

DISCUSSION

Masking has recently been examined in 3 rodent species in which individual differences in activity patterns, as well as in their degree of plasticity, are pronounced. In all of these, the unstriped Nile grass rat, the Mongolian gerbil, and the degus, masking was associated in predictable

ways with the overall activity pattern that the individuals exhibited at the time of testing (Hagenauer and Lee, 2008; Redlin and Mrosovsky, 2004; Vivanco et al., 2009; Weinert et al., 2007). This was the case for some but not all features of masking seen here in spiny mice.

Dark Pulses from ZT 2–5

A pulse of darkness during the day stimulated an increase in activity of both the common and the

when they are moved from the field to the lab.

Light Pulses from ZT 14-17

Responses of common spiny mice to light pulses during the dark phase of the LD cycle were again consistent and predictable. This was not the case for golden spiny mice, in which activity decreased, increased, or remained unchanged during the light pulse. Here, golden spiny mice were very different from 3 other species in which change from diurnality to nocturnality can occur. When Nile rats and Mongolian gerbils are nocturnal, their activity is consistently suppressed by light pulses during the dark period (Redlin and Mrosovsky, 2004; Weinert et al., 2007), and there is indirect evidence that this is also the case in degus

(Vivanco et al., 2009). Thus, golden spiny mice are highly unusual in that when they are night active, they do not show masking responses to light that are characteristic of nocturnality in other species.

LD 3.5:3.5 Cycles

The use of an ultradian LD cycle to which circadian rhythms cannot entrain should reveal the interacting influences of circadian systems and masking on the temporal organization of behavior (Redlin and Mrosovsky, 1999). To our knowledge, this protocol has not been used with grass rats or gerbils, but it has provided evidence that masking responses are different

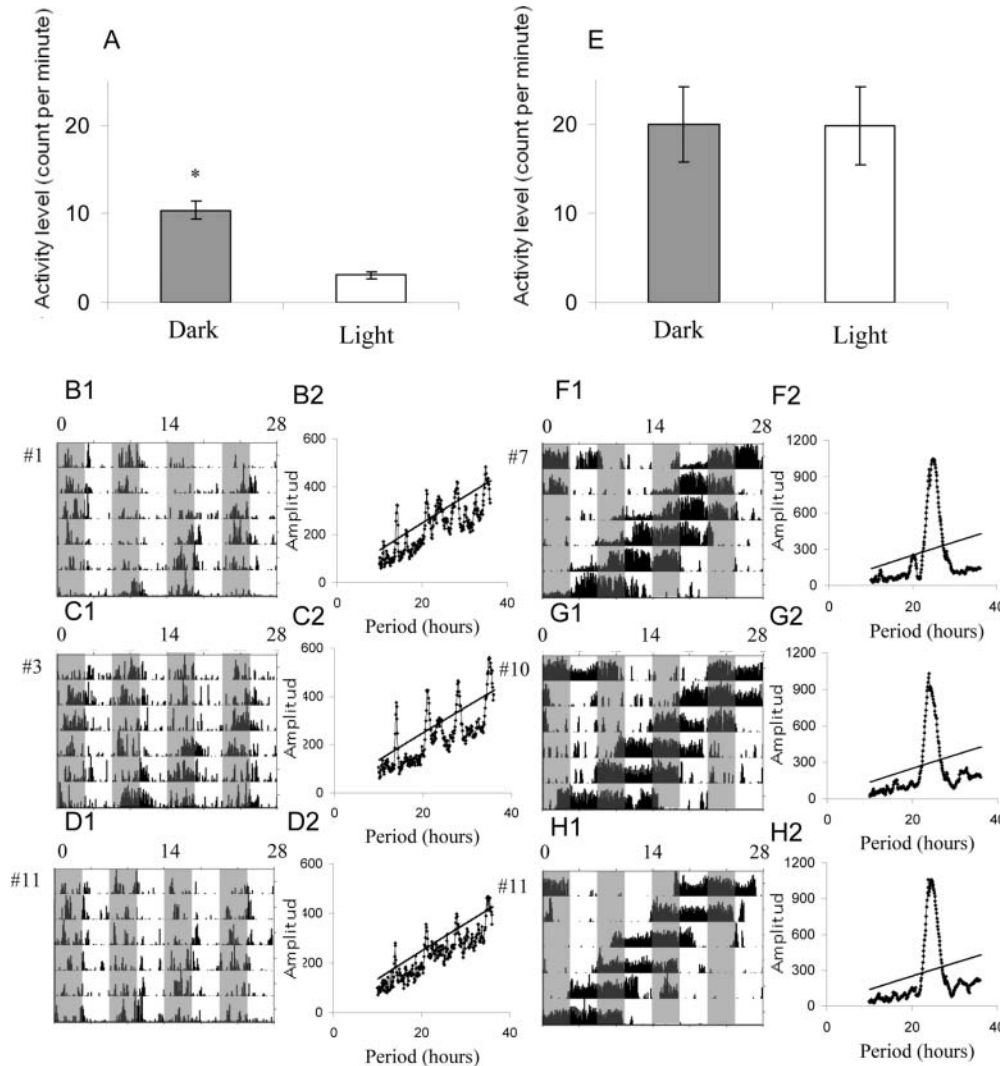


Figure 3. Activity levels (counts per minute, mean $\pm$ SE) during the dark (gray) and light (white) phases in (A) common ($n = 11$) and (E) golden ($n = 12$) spiny mice kept under 3.5:3.5 LD cycle, as well as representative actograms (28-h plots of common spiny mice [B1,C1,D1] and golden spiny mice [F1,G1,H1]; gray background represents the dark hours) with corresponding periodograms of the same individuals (common spiny mice: B2,C2,D2; golden spiny mice: F2,G2,H2). * $p < 0.0001$.

golden spiny mice (Fig. 1). This result is consistent with predictions for common spiny mice, but the story is more complicated for golden spiny mice as these animals are diurnal in their natural habitat. There are, however, some similarities between the patterns seen here and those apparent in captive Nile grass rats and Mongolian gerbils, species in which a switch to nocturnality can be induced by access to a running wheel. When this switch occurs, these animals begin to respond to dark pulses during the day with an increase in activity (Redlin and Mrosovsky, 2004; Weinert et al., 2007). Changing patterns of masking may contribute to the emergence of nocturnality in these animals, as well as in golden spiny mice,

in day- and night-active degus (Refinetti, 2006). Common spiny mice exhibited clear ultradian rhythms with peaks in activity during the dark phase of the ultradian cycle, a result consistent with those seen in the first 2 masking protocols. In golden spiny mice, by contrast, there were no ultradian rhythms, and there was no preference for either phase of the short LD cycle (Fig. 3). Thus, the absence of responsiveness to light was apparent in both experiments 2 and 3, whereas the presence of responsiveness to darkness was evident in experiment 1 but not experiment 3. The reasons for this paradox are unclear, but it provides further evidence that masking in these animals does not fit traditional expectations for either nocturnal or diurnal species.

Conclusions

In the present study, direct assessment of masking using identical protocols in 2 closely related species with very different activity rhythms revealed patterns that were highly predictable in some ways but quite novel in others. The common spiny mice behaved exactly as expected for a nocturnal rodent, indicating that traits associated with masking do not distinguish the genus *Acomys* from other mammalian taxa. However, within this genus, the golden spiny mice presented a more complex picture and did not match expectations for either a nocturnal or a diurnal species, nor did they conform to patterns of masking seen in other rodents with flexible activity patterns.

The existence of 2 closely related species with such marked differences in responsiveness to light raises the question of where within the nervous system these differences might originate. Possibilities include the melanopsin-containing retinal ganglion cells and the SCN to which they project, as both are important for maintenance of normal patterns of masking (e.g., Göz et al., 2008; Redlin, 2001). Brain regions that receive input from these 2 structures and play roles in regulation of sleep and arousal, such as the ventrolateral preoptic area and the lateral hypothalamus, are also possibilities. Investigation of these issues in golden and common spiny mice, perhaps with the use of cFos, might lead to a better understanding of masking more generally.

The current data provide further support for the hypothesis that golden spiny mice are in an intermediate state of adaptation representative of what can occur during evolutionary transitions from nocturnality to diurnality (Cohen and Kronfeld-Schor, 2006; Kronfeld-Schor and Dayan, 2003, 2008). That is, the different features of masking

may represent new examples of the unique constellation of adaptations these animals have, some supporting daytime and others nighttime activity (reviewed by Kronfeld-Schor and Dayan, 2008). It should also be noted that the resistance to inhibitory effects of light on activity seen in the present study is likely to have been essential for golden spiny mice to move into a diurnal niche. In fact, elimination of this form of masking may have occurred very early in every evolutionary transition from nocturnality to diurnality.

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